DRUGS · ANTIFUNGALS AND ANTIVIRALS

Aciclovir and valaciclovir

Aciclovir and its prodrug valaciclovir inhibit the DNA polymerase of herpes simplex and varicella zoster viruses. In ENT they are used for herpes zoster oticus (Ramsay Hunt syndrome), with steroids; for severe or recurrent herpes labialis and herpetic gingivostomatitis; for herpes zoster of the trigeminal or cervical dermatomes; and, though the evidence is weak, sometimes added to steroids in severe Bell's palsy.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

ACICLOVIR AND VALACICLOVIR

	SUMMARY
Class	Nucleoside analogue antivirals (famciclovir is similar)
How they work	Phosphorylated by viral thymidine kinase, then by host kinases to aciclovir triphosphate, which inhibits viral DNA polymerase and terminates the DNA chain; active only in infected cells
ENT uses	Ramsay Hunt syndrome (with prednisolone); herpes zoster of the head and neck; severe primary herpetic gingivostomatitis; herpes simplex in the immunocompromised; severe Bell's palsy (adjunct, weak evidence); herpes simplex encephalitis (intravenous)
Adult dose	Zoster: aciclovir 800 mg five times daily or valaciclovir 1 g three times daily for 7 days, started within 72 hours; intravenous aciclovir 10 mg/kg every 8 hours for severe, disseminated, central or immunocompromised infection
Child dose	Aciclovir by age and weight (for example 20 mg/kg four times daily for zoster in children over 2, maximum 800 mg)
Pregnancy	Aciclovir has extensive safety data

2 CHAPTER TWO Using them in ENT

Indications.

2.1 Ramsay Hunt syndrome

Facial palsy with a vesicular eruption in the ear canal, concha or palate, often with hearing loss, tinnitus and vertigo. Start an antiviral and **prednisolone** within 72 hours, protect the eye, and arrange audiometry; recovery is worse than in Bell's palsy, so counsel accordingly.

2.2 Bell's palsy

Steroids alone are the evidence-based treatment; adding an antiviral may slightly improve outcomes in severe palsy, but the evidence is low quality.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Nausea, headache, rash; with intravenous aciclovir, **crystal nephropathy** (hydrate well, infuse slowly) and neurotoxicity (confusion, tremor) in renal impairment.

3.2 Interactions

Other nephrotoxins; probenecid raises levels; mycophenolate (both levels raised).

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Mechanism and selectivity (viral thymidine kinase).
- Treatment of **Ramsay Hunt syndrome**.
- Dose adjustment and hydration in renal impairment.

4.2 Common pitfalls

COMMON PITFALLS

- Missing a vesicle in the ear canal in a "Bell's palsy".
- Intravenous aciclovir without hydration in an elderly patient.

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